notebook2

The purpose of this notebook is to record my decision making during the model building process. I will start with a simple model and work up to the final model, that way I can check computation and model fit along the way.

Load in packages.

```
library(cmdstanr)
library(dplyr)
library(tidyr)
library(ggplot2)
library(bayesplot)
library(posterior)
```

Overall relative growth rate

First, the simplest model is just fitting a single intercept and slope to the entire data set. The plants are growing exponentially during this life-stage: $s_t = s_0 e^{rt}$. Here s_t is size at time t, and s_0 is the original size. r is the growth rate, here relative growth rate, and t is the time since germination of the seedling. By taking the log of this we get: $\log(s_t) = \log(s_0) + rt$ which we can fit using a normal linear regression.

Here's the model:

$$
\begin{aligned}
\log(s_{it}) &\sim \text{Normal}(\mu_{it}, \sigma) \\
\mu_{it} &= \log(s_0) + rgr \times t \\
\alpha &\sim \text{Normal}(0, 1) \\
rgr &\sim \text{Normal}(0.24, 0.1) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
\tag{1}
$$

```
data{
  int N; // number of observations
  vector[N] l_size; // size observations
  vector[N] age; // age of observations
}
parameters{
  real l_size0; // log size at time 0
  real rgr; // relative growth rate
  real<lower=0> sigma; // standard deviation of errors
}
model{
  // priors
  l_size0 ~ normal(0, 1);
  rgr ~ normal(0.24, 0.1);
  sigma ~ exponential(1);

  // likelihood
  l_size ~ normal(l_size0 + rgr*age, sigma);
}
```

Now use Equation 1.